Machine Learning: Clustering & Classification

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References:

- Machine Learning with R. Lantz, 4th edition, 2023
- An Introduction to Statistical Learning with Applications in .R James, Witten, Hastie, Tibshirani; 2nd edition, 2023

https://hastie.su.domains/ISLR2/ISLRv2_corrected_June_2023.pdf.download.html

I. Some Aspects of Machine Learning (ML)

A. Definition

A set of methods for automatically detecting patterns in data; the uncovered patterns are used in decision making to predict future outcomes and plan additional data collection.
K.P. Murphy, Machine Learning. 2012.

B. Some applications

- Identification of spam messages in email
- Segmentation of customer behavior for targeted advertising
- Detection of fraudulent credit card transactions
- Optimization of energy use in homes and buildings
- Discovery of genetic sequences linked to disease
- Algorithms for self driving cars & autopiloting drones

C. Types of learning algorithms

Types of tasks:

- Classification – assigning an object to a predefined group/class
- Numeric prediction – determining an output value from input data
- Pattern discovery – identify occurrences that appear repeatedly
- Clustering – identifying groups within a data set
- Generation – create new data samples

Types of learning:

- Supervised – learning from past data with known results
- Unsupervised learning – learning from past data w/o results
- Reinforcement learning – learning from past data w/ feedback

Some ML methods based on the algorithm type and learning task

Algorithm	Learning task	Chapter
Supervised learning algorithms		
k-nearest neighbors	Classification	*Chapter 3*
Naive Bayes	Classification	*Chapter 4*
Decision trees	Classification	*Chapter 5*
Classification rule learners	Classification	*Chapter 5*
Linear regression	Numeric prediction	*Chapter 6*
Regression trees	Numeric prediction	*Chapter 6*
Model trees	Numeric prediction	*Chapter 6*
Neural networks	Dual use	*Chapter 7*
Support vector machines	Dual use	*Chapter 7*
Unsupervised learning algorithms		
Association rules	Pattern detection	*Chapter 8*
k-means clustering	Clustering	*Chapter 9*
Meta-learning algorithms		
Bagging	Dual use	*Chapter 11*
Boosting	Dual use	*Chapter 11*
Random forests	Dual use	*Chapter 11*

From: B. Lantz, Machine Learning with R. 2019.

II. The Clustering and Classification Problems

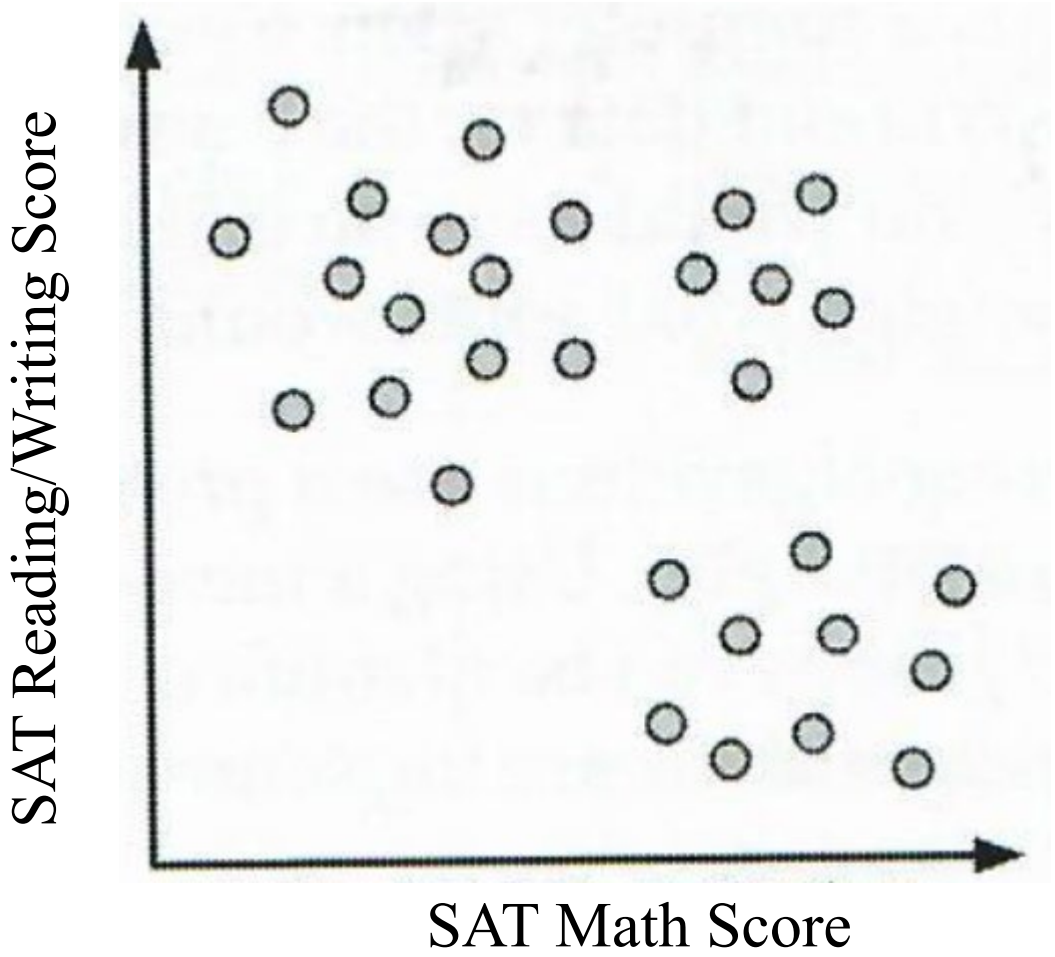
A. Clustering: problem & illustration

Clustering Problem:

- Given: A vector of features/attributes for each element in set of data (training data).
- Problem: How can we group the training set data into several cohesive clusters?
- Approach: The data within a cluster are grouped based on their similarity; the clusters are defined by their dissimilarity.
- Purpose: Uncover previously undetected relations in a set of data.
- Examples:
  - Segmenting customers into groups with similar buying patterns
  - Image compression by grouping similar features into a smaller number of homogeneous categories (e.g., Principal Component Analysis)
  - Detecting anomalous network intrusions that fall outside of known clusters

Clustering Illustration SAT scores

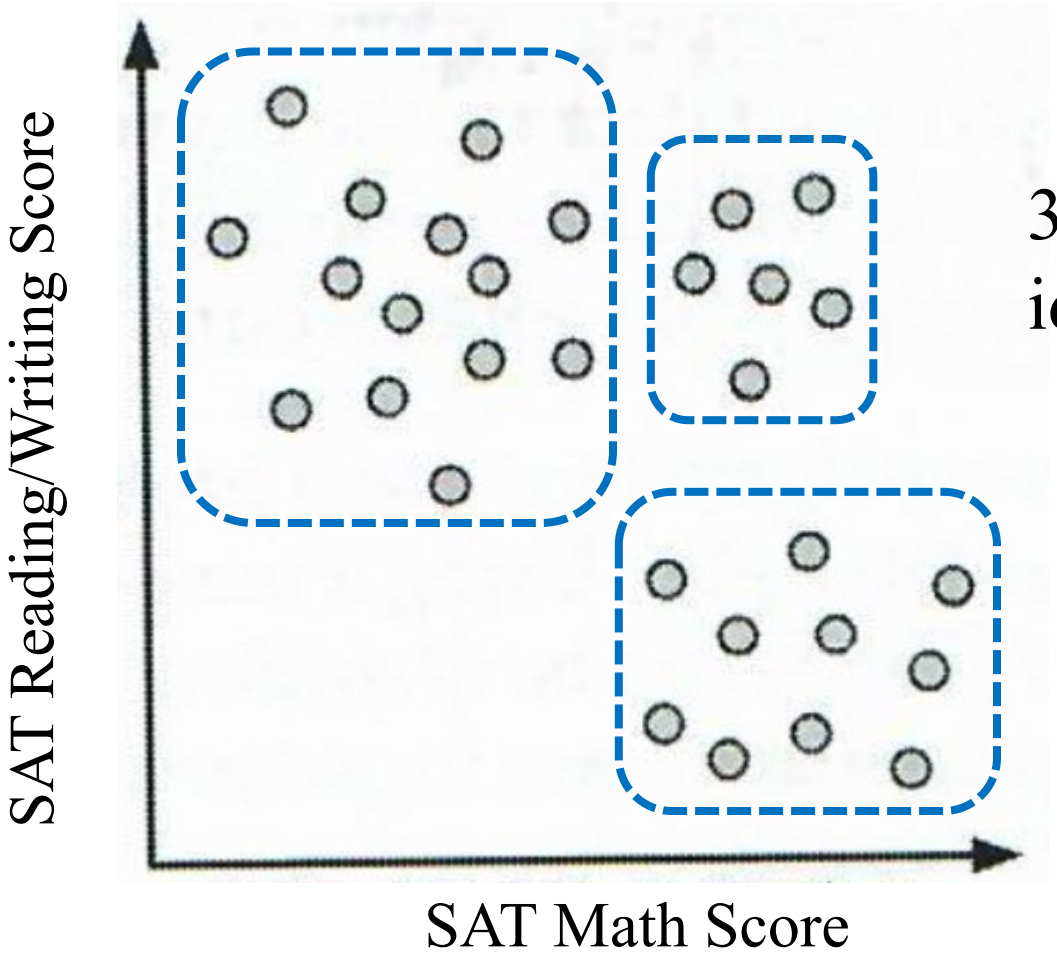
Feature space plot



Can you visually identify clusters/groupings?

Clustering Illustration SAT scores

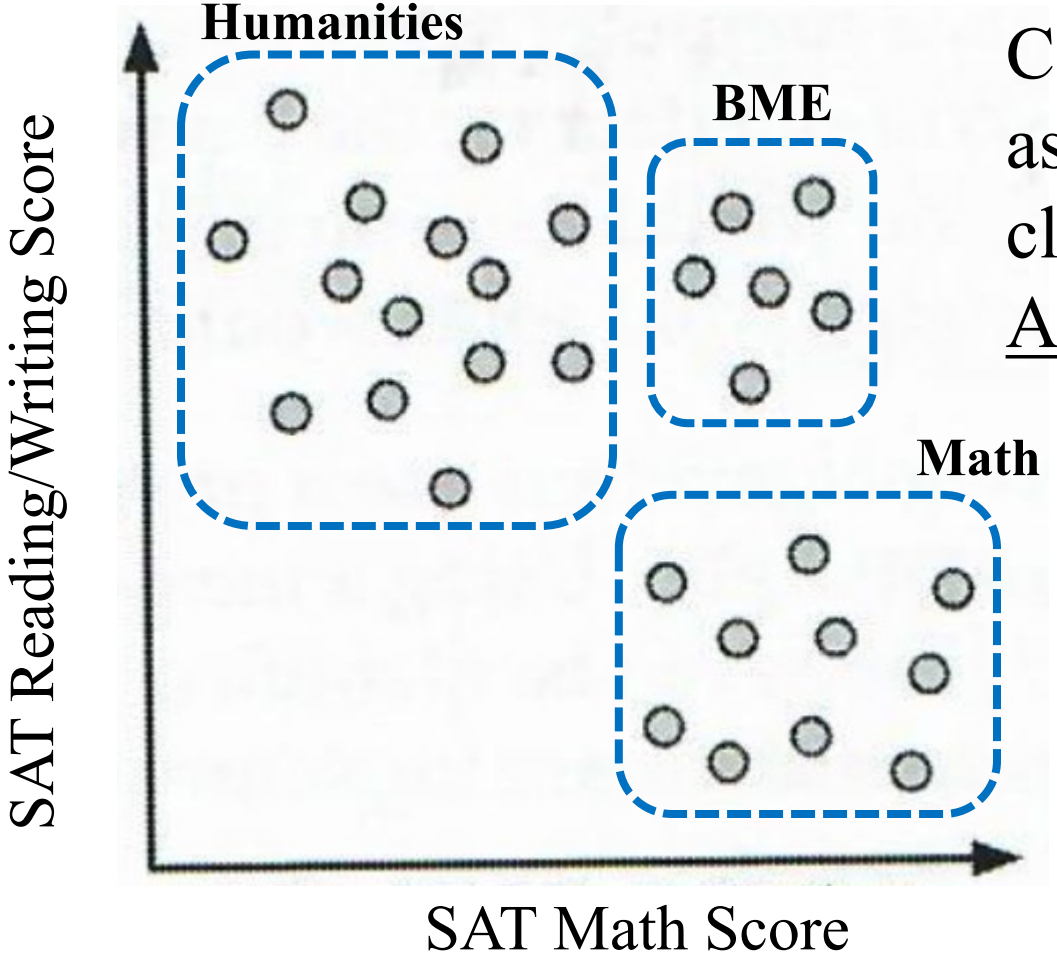
Feature space plot



3 clusters/groups identified

Clustering Illustration SAT scores

Feature space plot



Class designation assigned for each cluster/group – Accepted Major

Associate clusters with other collected information about applicants:
e.g., country, US state, size of high school, accepted major, etc.

B. Classification: problem & illustration

Classification Problem:

- Given: A vector of features/attributes for each element in a set of data, along with a set of predetermined groups/classes for each element in the data set.
- Problem: What rule(s) can we use to assign a new object to one of the groups/classes?
- Approach: The new object is assigned to the group/class that has an object or objects that it is most similar to.
- Purpose: Assigning an object to one of a set of predefined groups.
- Examples:
  - Detecting spam email – spam versus ham
  - Optical character recognition (e.g., zip code on envelope)
  - Detecting faces in images
  - Predicting music preferences
  - Assigning a phenotype given a genotype

Solving the Classification Problem (four main types of algorithms):

- Frequency Table:
  - ZeroR
  - OneR
  - Naïve Bayesian
  - Decision Tree (DT) BME 210
- Covariance Matrix:
  - Discriminant Analysis
  - Logistic Regression BME 423
- Similarity Functions:
  - K-Nearest Neighbors (KNN) BME 210
- Others:
  - Neural Networks BME 423
  - Support Vector Machines

Classification Illustration

attributes/features

group/class

Ingredient	Sweetness	Crunchiness	Food type
Apple	10	9	Fruit
Bacon	1	4	Protein
Banana	10	1	Fruit
Carrot	7	10	Vegetable
Celery	3	10	Vegetable
Cheese	1	1	Protein

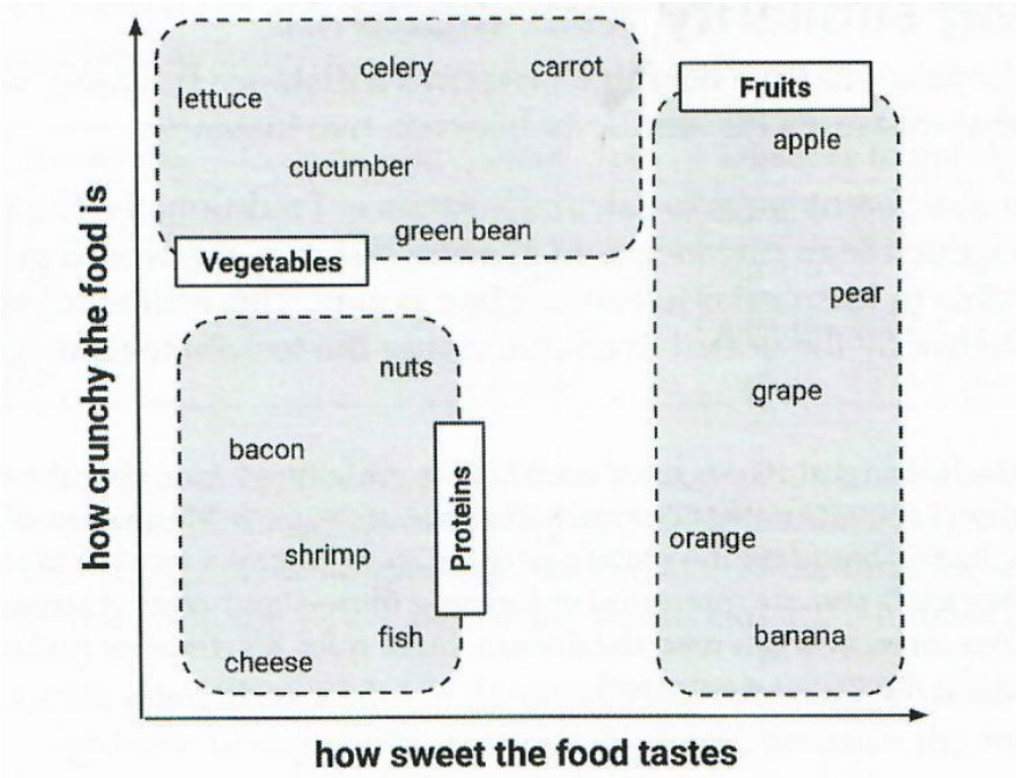


Figure 3.2: Foods that are similarly classified tend to have similar attributes

Classification Illustration

How do we classify tomatoes?

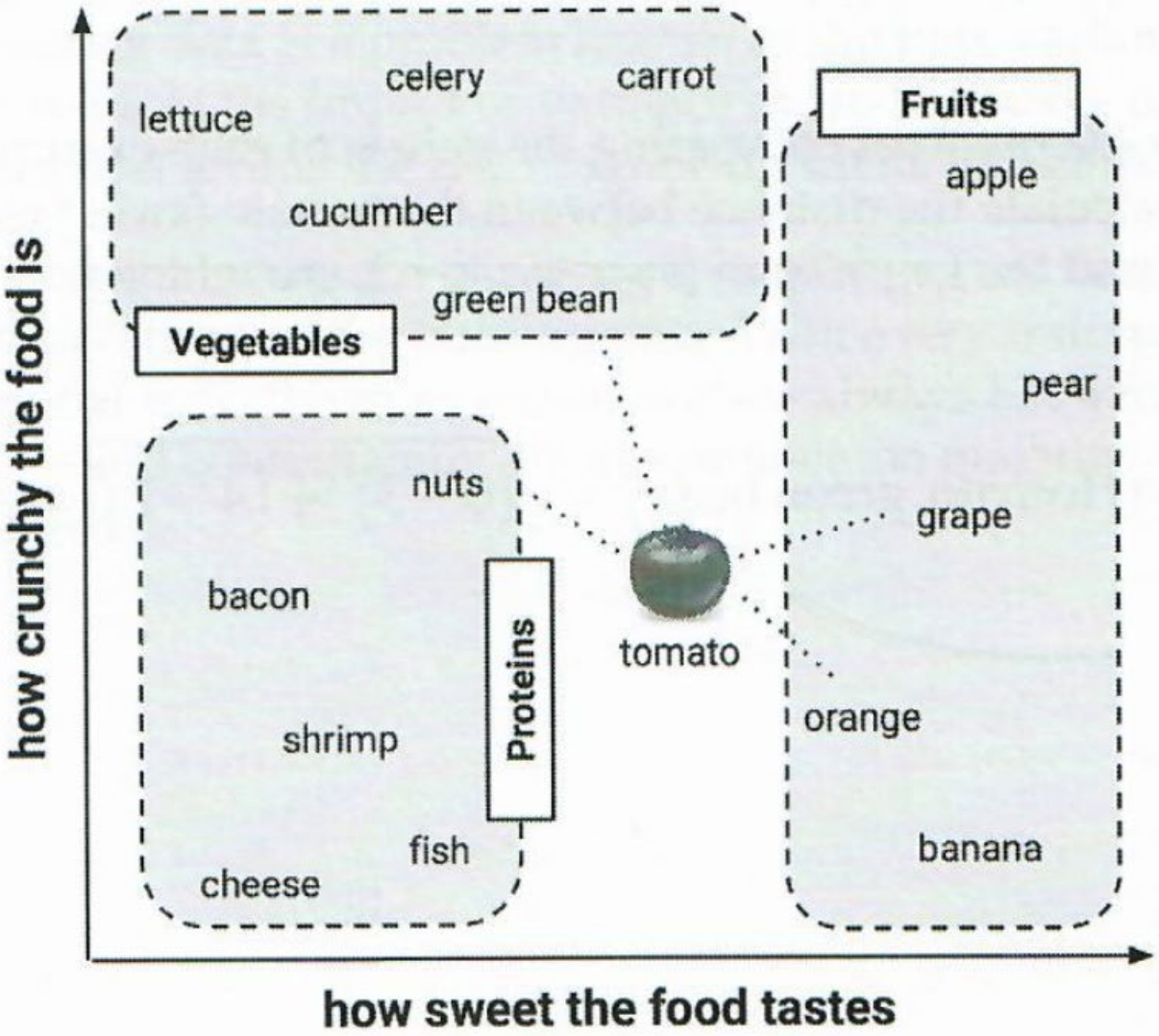


Figure 3.3: The tomato's nearest neighbors provide insight into whether it is a fruit or vegetable

III. k-Means Clustering

A. Notation

x – vector of features / attributes (dim D)

$x_i,\ i = 1,\ldots,n$ – vector of features / attributes for each of the entries (n) in the data set (dim D)

X – matrix of features/attributes dim ($n \times D$)

k – number of clusters

$y_i,\ i = 1,\ldots,n$ – cluster label ($1,\ldots, k$) for each of the n entries in the data set

$c_j,\ j = 1,\ldots,k$ – centroid of each cluster (dim D)

C – matrix of centroids of each cluster dim ($k \times D$)

B. Distance Measures

Similarity or dissimilarity can be quantified using distance measures.

As an example, consider two points, *q* and *r*, in two-dimensional space:

$$q = [q_1 \quad q_2]; \quad r = [r_1 \quad r_2]$$

The Euclidian distance between these two points is defined as:

$$d(q,r) = \sqrt{(q_1 - r_1)^2 + (q_2 - r_2)^2}$$ (*d* has the same units as *q* and *r*)

When $d(q,r)$ is small, the points are considered close or similar.

When $d(q,r)$ is large, the points are considered far apart or dissimilar.

Some distance measures in m-space:

Consider two points, q and r, in m-dimensional space:

$$q = [q_1 \quad \cdots \quad q_m]; \qquad r = [r_1 \quad \cdots \quad r_m]$$

The following distance measures can be defined:

Euclidean $\quad d(q,r) = \sqrt{\sum_{i=1}^{m}(q_i - r_i)^2} \quad \|q-r\|_2 \quad \ell_2$ norm

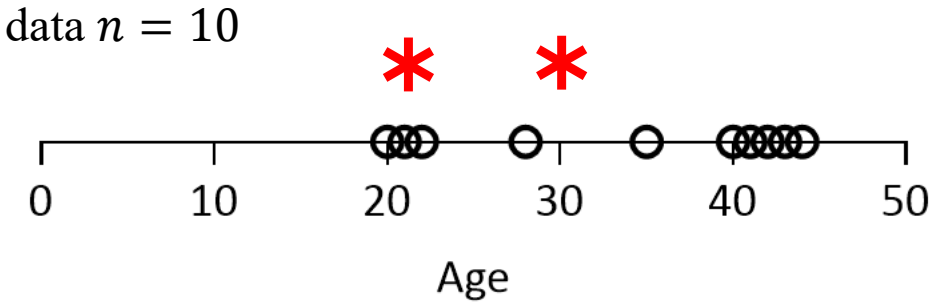
Manhattan $\quad d(q,r) = \sum_{i=1}^{m}|q_i - r_i| \quad \|q-r\|_1 \quad \ell_1$ norm

Minkowski $\quad d(q,r) = \left(\sum_{i=1}^{m}|q_i - r_i|^p\right)^{1/p} \quad \|q-r\|_p \quad \ell_p$ norm

norm: a function that assigns a strictly positive length or size to each vector in a vector space

C. One-dimensional example – distribution of ages

https://www.saedsayad.com/clustering_kmeans.htm



of clusters: $k = 2$

Initial centroids:

$c_1 = 21 \quad c_2 = 30$

Iteration #1

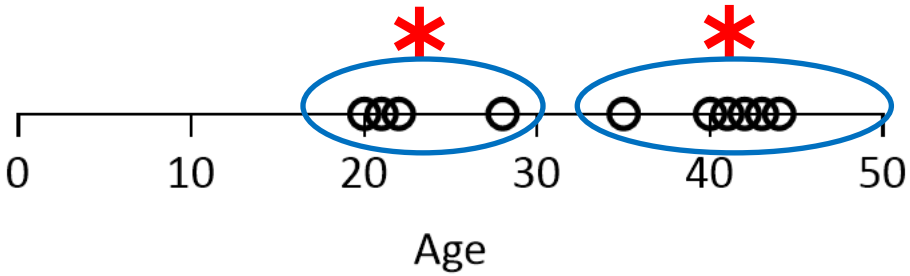
x_i	c_1	c_2	Distance to c1	Distance c2	Nearest Cluster	New Centroid
20	21	30	1	10	1	
21	21	30	0	9	1	$\frac{1}{3}\sum_{i=1}^{3} x_i \rightarrow 21$
22	21	30	1	8	1	
28	21	30	7	2	2	
35	21	30	14	5	2	
40	21	30	19	10	2	$\frac{1}{7}\sum_{i=4}^{10} x_i \rightarrow 39$
41	21	30	20	11	2	
42	21	30	21	12	2	
43	21	30	22	13	2	
44	21	30	23	14	2	

C. One-dimensional example – distribution of ages

Centroids from iter. # 1

Iteration #2

x_i	c_1	c_2	Distance to c1	Distance c2	Nearest Cluster	New Centroid
20	21	39	1	19	1	$\frac{1}{4}\sum_{i=1}^{4} x_i \rightarrow 22.75$
21	21	39	0	18	1	
22	21	39	1	17	1	
28	21	39	7	11	1	
35	21	39	14	4	2	$\frac{1}{6}\sum_{i=5}^{10} x_i \rightarrow 40.83333$
40	21	39	19	1	2	
41	21	39	20	2	2	
42	21	39	21	3	2	
43	21	39	22	4	2	
44	21	39	23	5	2	

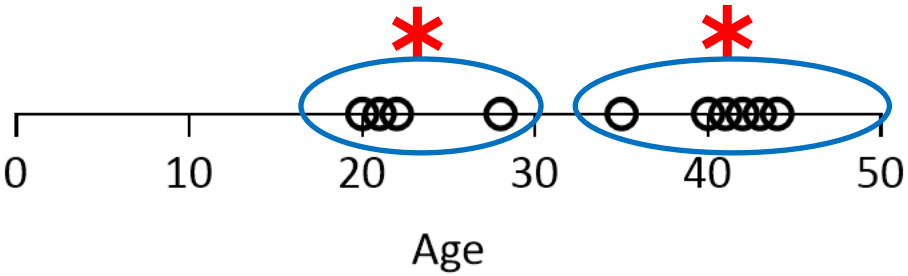


C. One-dimensional example – distribution of ages

Centroids from iter. # 2

Iteration #3

x_i	c_1	c_2	Distance to c1	Distance c2	Nearest Cluster	New Centroid
20	22.75	40.8333	2.75	20.83333	1	$\frac{1}{4}\sum_{i=1}^{4} x_i \rightarrow 22.75$
21	22.75	40.8333	1.75	19.83333	1	
22	22.75	40.8333	0.75	18.83333	1	
28	22.75	40.8333	5.25	12.83333	1	
35	22.75	40.8333	12.25	5.83333	2	$\frac{1}{6}\sum_{i=5}^{10} x_i \rightarrow$
40	22.75	40.8333	17.25	0.83333	2	
41	22.75	40.8333	18.25	0.16667	2	40.83333
42	22.75	40.8333	19.25	1.16667	2	
43	22.75	40.8333	20.25	2.16667	2	
44	22.75	40.8333	21.25	3.16667	2	



Define two categories:
- young
- old

D. Algorithm

Given this illustration, we now define the k-Means algorithm

- The k-Means algorithm assigns each of n entries in the data set to one of k clusters (k is selected *a priori*).
- The objective is to minimize the differences within each cluster and maximize the differences between clusters.

Algorithm (the formal derivation is omitted)

1. Select k points at random from the data set to serve as initial centroids
2. Assign each entry to its nearest cluster center based on its Euclidean distance:
$$y_i = \arg\min_j \left\| x_i - c_j \right\|_2, \qquad i = 1,...,n, \quad j = 1,...,k$$
3. Calculate the centroid of all entries within each cluster:
$$c_j = \frac{1}{n_j} \sum_{i:y_i=j} x_i, \quad j = 1,...,k$$
4. Repeat steps 2 and 3 until no changes in cluster assignments

C. One-dimensional example – distribution of ages

$$y_i = \arg\min_j \left\| x_i - c_j \right\|_2, \qquad i = 1,\ldots,n, \quad j = 1,\ldots,k$$

x_i	c_1	c_2	Distance to c1	Distance c2	Nearest Cluster	New Centroid
20	21	30	1	10	1	
21	21	30	0	9	1	$\frac{1}{3}\sum_{i=1}^{3} x_i \rightarrow 21$
22	21	30	1	8	1	
28	21	30	7	2	2	
35	21	30	14	5	2	
40	21	30	19	10	2	$\frac{1}{7}\sum_{i=4}^{10} x_i \rightarrow 39$
41	21	30	20	11	2	
42	21	30	21	12	2	
43	21	30	22	13	2	
44	21	30	23	14	2	

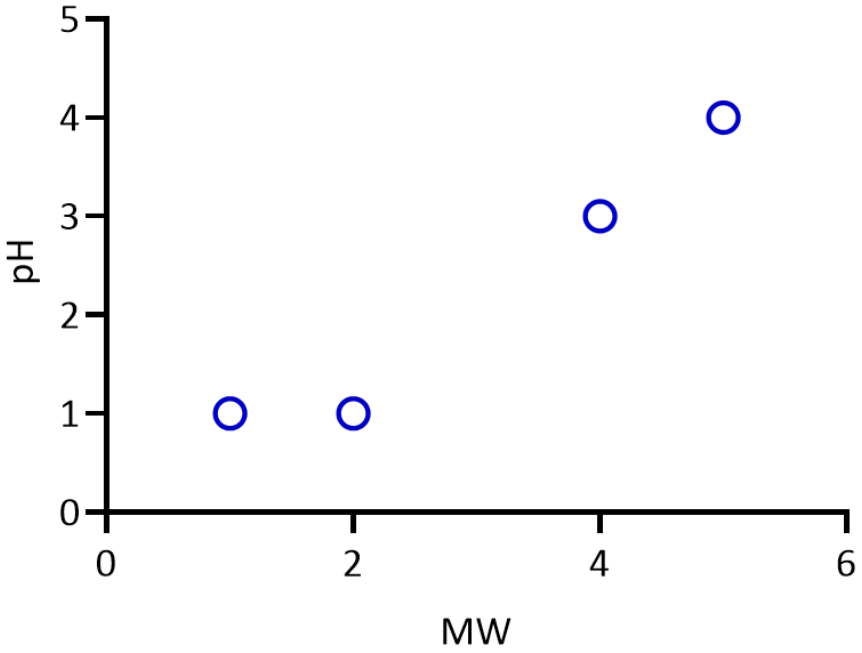
D. Two-dimensional example – medicine MW & pH

The action of drugs can sometimes be group based on their physiochemical characteristics.

The table below shows 4 medicines with different molecular weights (MW) and pH values (for this illustration, MW and pH are represented on arbitrary scales).

Medicine	Feature 1 (MW)	Feature 2 (pH)
1	1	1
2	2	1
3	4	3
4	5	4

Spreadsheet of features/attributes



1. Select 2 points at random to serve as initial centroids

$$c_1 = (1, 1) \quad c_2 = (2, 1) \qquad (k = 2)$$

2. Assign each entry to its nearest cluster centroid (Euclidean distance)

distance $\quad d_{ij} = \left\| x_i - c_j \right\|_2, \qquad i = 1,\ldots,4, \quad j = 1,\ldots,2$

cluster # $\quad y_i = \arg\min_j \left\| x_i - c_j \right\|_2, \qquad i = 1,\ldots,4, \quad j = 1,\ldots,2$

Medicine	*xi*	*c1*	*c2*	Distance to c1	Distance to c2	Cluster assignment yi	New centroid
1	(1, 1)	(1, 1)	(2, 1)	0	1	1	
2	(2, 1)	(1, 1)	(2, 1)	1	0	2	
3	(4, 3)	(1, 1)	(2, 1)	3.61	2.83	2	
4	(5, 4)	(1, 1)	(2, 1)	5	4.24	2	

distance

$$d_{11} = \sqrt{((1-1)^2 + (1-1)^2}$$

$$d_{12} = \sqrt{((1-2)^2 + (1-1)^2}$$

cluster #

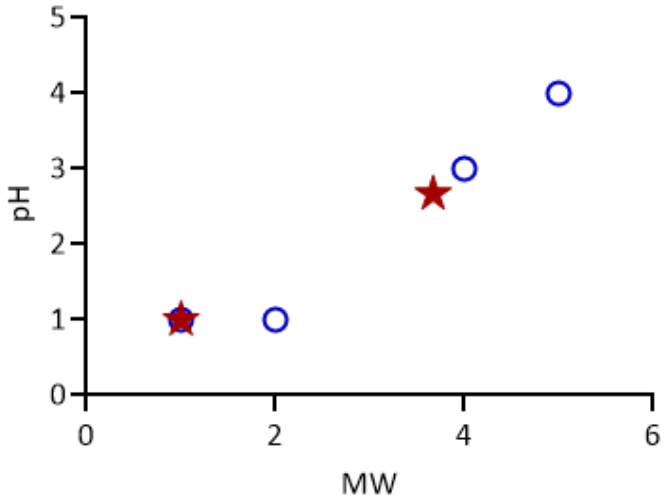
$$y_1 = \arg\min_j (d_{1j=1}, d_{1j=2}) = 1$$

3. Calculate the centroid of all entries within each cluster:

$$c_j = \frac{1}{n_j} \sum_{i:y_i=j} x_i, \quad j = 1,\ldots,k$$

Medicine	*xi*	*c1*	*c2*	Distance to c1	Distance to c2	Cluster assignment yi	New centroid
1	(1, 1)	(1, 1)	(2, 1)	0	1	1	(1,1)
2	(2, 1)	(1, 1)	(2, 1)	1	0	2	
3	(4, 3)	(1, 1)	(2, 1)	3.61	2.83	2	(3.67, 2.67)
4	(5, 4)	(1, 1)	(2, 1)	5	4.24	2	

$$c_2 = \frac{1}{3} \sum_{i=2}^{4} x_i$$



Iteration # 2

2. Assign each entry to its nearest cluster centroid (Euclidean distance)

distance $d_{ij} = \|x_i - c_j\|_2, \quad i = 1,\ldots,4, \quad j = 1,\ldots,2$

cluster # $y_i = \arg\min_j \|x_i - c_j\|_2, \quad i = 1,\ldots,4, \quad j = 1,\ldots,2$

Centroids from iter. # 1

Medicine	*xi*	*c1*	*c2*	Distance to c1	Distance to c2	Cluster assignment yi	New centroid
1	(1, 1)	(1, 1)	(3.67, 2.67)	0	3.14	1	
2	(2, 1)	(1, 1)	(3.67, 2.67)	1	2.36	1	
3	(4, 3)	(1, 1)	(3.67, 2.67)	3.61	0.4715	2	
4	(5, 4)	(1, 1)	(3.67, 2.67)	5	1.89	2	

distance

$$d_{11} = \sqrt{((1-1)^2 + (1-1)^2}$$

$$d_{12} = \sqrt{((1-3.67)^2 + (1-2.67)^2}$$

cluster #

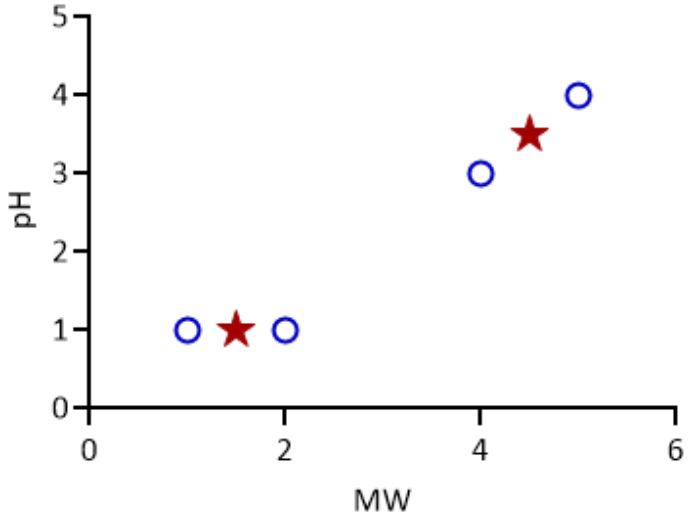
$$y_1 = \arg\min_j (d_{1j=1}, d_{1j=2}) = 1$$

Iteration # 2

3. Calculate the centroid of all entries within each cluster:

$$c_j = \frac{1}{n_j} \sum_{i:y_i=j} x_i, \quad j = 1,\ldots,k$$

Medicine	*xi*	*c1*	*c2*	Distance to c1	Distance to c2	Cluster assignment yi	New centroid
1	(1, 1)	(1, 1)	(3.67, 2.67)	0	3.14	1	(1.5,1)
2	(2, 1)	(1, 1)	(3.67, 2.67)	1	2.36	1	
3	(4, 3)	(1, 1)	(3.67, 2.67)	3.61	0.4715	2	(4.5,3.5)
4	(5, 4)	(1, 1)	(3.67, 2.67)	5	1.89	2	



$$c_2 = \frac{1}{2} \sum_{i=3}^{4} x_i$$

Session 22 Lecture Exercise

Systolic blood pressure (SBP; mmHg) and C-reactive protein (CRP; mg/L) were measured in 3 adult subjects. The data were then standardized to avoid any issues associated with differences in scale; the scaled data are unitless. Based on the resulting data listed in the table below, perform a single iteration of the k-Means algorithm with $k = 2$. Use the Euclidian distance formula for your calculations.

Use the following two initial centroids for your calculations:
C1: $\text{SBP} = 2, \text{CRP} = 2$
C2: $\text{SBP} = 5, \text{CRP} = 5$

Patient	SBP	CRP
1	1	3
2	7	2
3	8	7

As part of your final answer, provide the cluster labels for each of the three data points AND the new centroids of the two updated clusters after this single iteration. Make sure to show your intermediate calculations.

F. The k-means function in R

`kmeans(X, centers)`

See R Help for full list of arguments

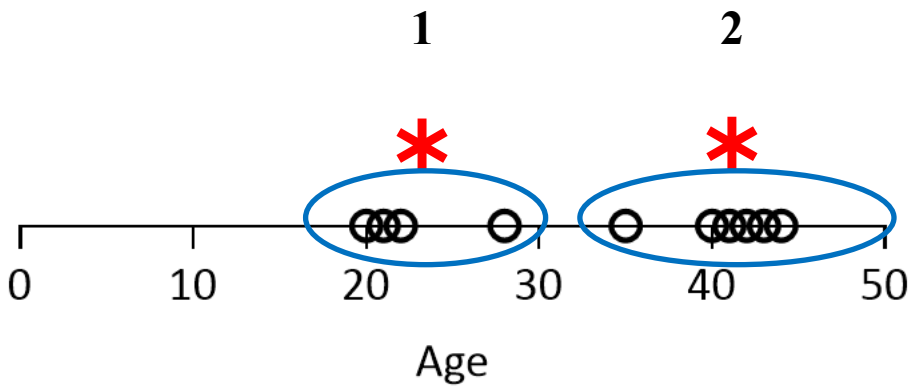
X – matrix of features/attribute dim nxD
centers – number of clusters (k)
or
coordinate matrix (C) of
cluster centers

Sample R code for 1 dimensional age example

```
age <- c(20, 21, 22, 28, 35, 40, 41, 42, 43, 44)
initial=c(21, 30)
clusterresults<-kmeans(age, initial)
fitted(clusterresults, method = c("centers", "classes"))
clusterresults$size
clusterresults$withinss
clusterresults$tot.withinss
clusterresults$betweenss
```

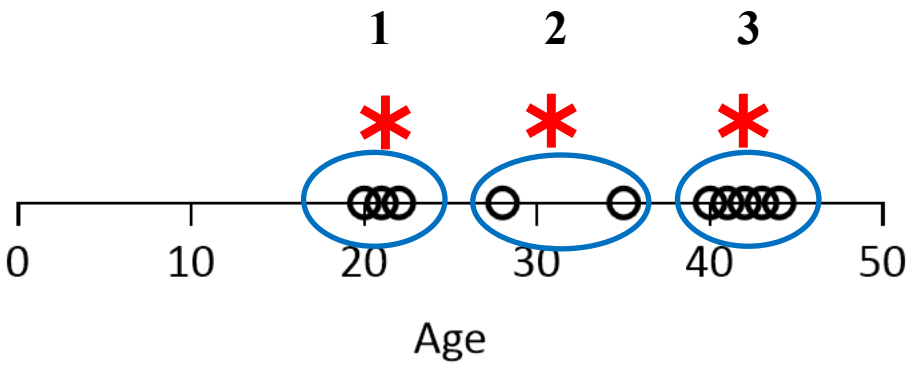
Results from sample R code for 1 dimensional age example: 2 clusters

```
> # 2 clusters
> initial=c(21, 30)
> clusterresults<-kmeans(age, initial)
> fitted(clusterresults)
      [,1]
1 22.75000
1 22.75000
1 22.75000
1 22.75000
2 40.83333
2 40.83333
2 40.83333
2 40.83333
2 40.83333
2 40.83333
> clusterresults$size
[1] 4 6
> clusterresults$withinss
[1] 38.75000 50.83333
> clusterresults$tot.withinss
[1] 89.58333
> clusterresults$betweenss
[1] 784.8167
```



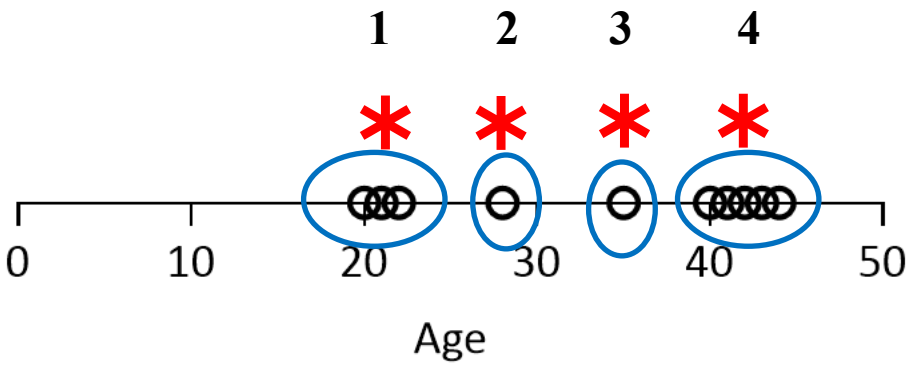
Results from sample R code for 1 dimensional age example: 3 clusters

```
> # 3 clusters
> initial=c(25, 35, 45)
> clusterresults<-kmeans(age, initial)
> fitted(clusterresults)
   [,1]
1 21.0
1 21.0
1 21.0
2 31.5
2 31.5
3 42.0
3 42.0
3 42.0
3 42.0
3 42.0
> clusterresults$size
[1] 3 2 5
> clusterresults$withinss
[1]  2.0 24.5 10.0
> clusterresults$tot.withinss
[1] 36.5
> clusterresults$betweenss
[1] 837.9
```



Results from sample R code for 1 dimensional age example: 4 clusters

```
> # 4 clusters
> initial=c(22, 30, 35, 42)
> clusterresults<-kmeans(age, initial)
> fitted(clusterresults)
   [,1]
1    21
1    21
1    21
2    28
3    35
4    42
4    42
4    42
4    42
4    42
4    42
> clusterresults$size
[1] 3 1 1 5
> clusterresults$withinss
[1]  2  0  0 10
> clusterresults$tot.withinss
[1] 12
> clusterresults$betweenss
[1] 862.4
```



G. Number of clusters

How do we decide on the appropriate number of clusters using k-Means?

The trade off between too few $(k \to 1)$ or too many $(k \to n)$ clusters is evident.

Many formal methods have been proposed and are used to select k.

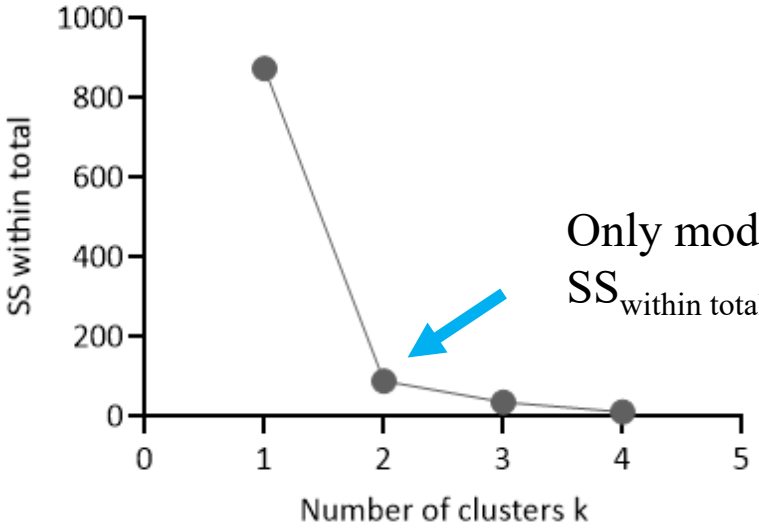
Conceptually they are based on how the heterogeneity (differences) within clusters changes with k. This can be quantified by calculating the within cluster sum of squared deviations from the centroid of the cluster.

Results for age example

$$SS_j = \sum_{i:y_i=j} \left(x_i - c_j\right)^2, \quad j = 1,\ldots,k$$

$$SS_{\text{within total}} = \sum_{j=1}^{k} SS_j$$

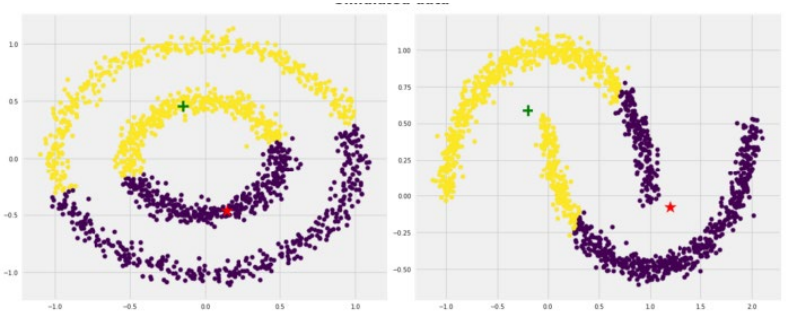
within-group/cluster heterogeneity



Only modest improvement in $SS_{\text{within total}}$ when $k > 2$

H. Limitations of k-Means clustering

- Not computationally practicable for big data sets
- Must specify number of clusters *a priori*
- Results may depend on selection of initial clusters
- Each factor has the same weight/contribution to the grouping process
- The Euclidean distance can be distorted by extreme points
- Does not work well if groups are not hyperellipsoids



k-Means cannot identify the correct clusters

There are many other clustering methods/algorithm that overcome these limitations

IV. K-Nearest Neighbors (K-NN) Classification

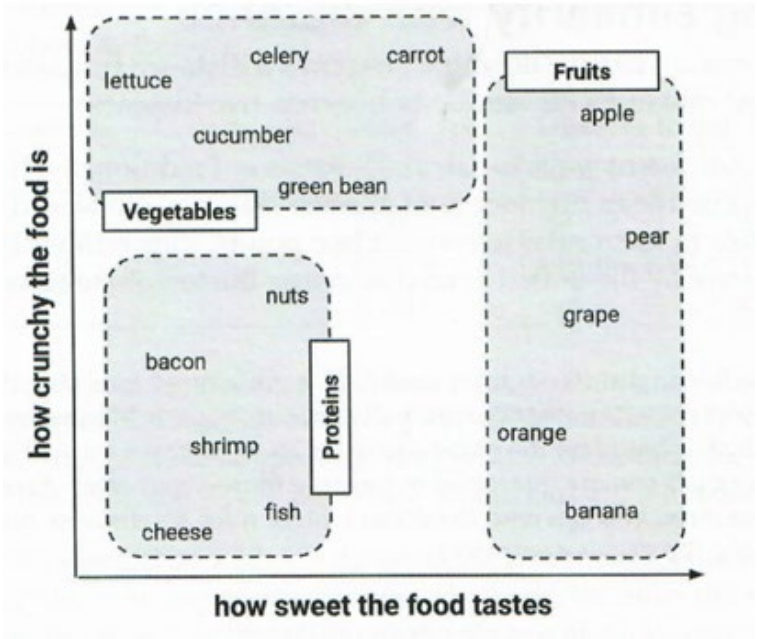
A. Classification Problem

- Given: A vector of features/attributes for each of the n objects
  in the set of data (input): $x_i,\ i = 1,...,n$ (dim D)
  and a set of class labels for each element
  in the data set (output): $y_i \in (1,...,L)$ (scalar)
  Training Set
- Problem: What rule(s) can we use to assign a new object to one of the groups/classes? Test Set
- Approach: The new object is assigned to the group/class that has an object or objects that it is most similar to.

Build model using Training Set data, then use model to classify Test Set data

Classification is an example of supervised learning, the most common type of ML in practice

B. K-Nearest Neighbors (K-NN) Classification – Illustrations

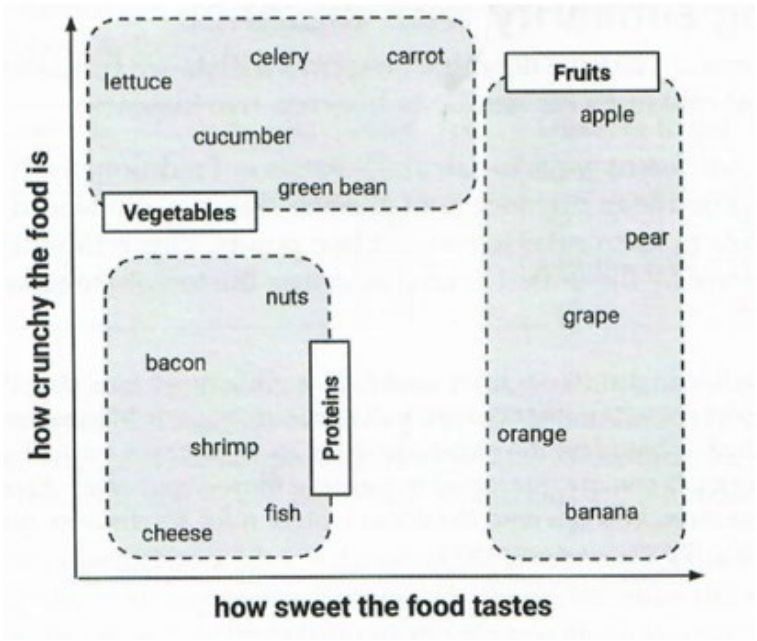


Training Set ($n = 15$) Attribute Vector ($D = 2$) Class Labels ($L = 3$)

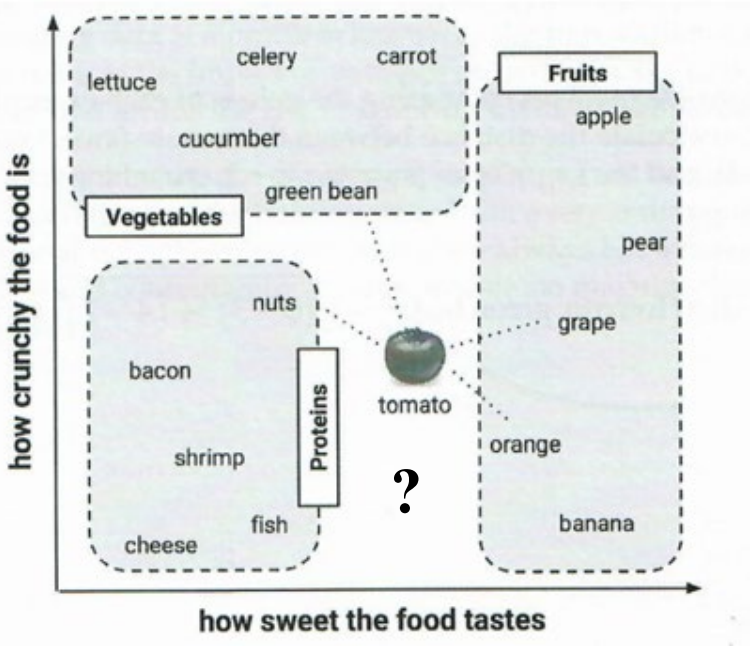
Ingredient	Sweetness	Crunchiness	Food type
Apple	10	9	Fruit
Bacon	1	4	Protein
Banana	10	1	Fruit
Carrot	7	10	Vegetable
Celery	3	10	Vegetable
Cheese	1	1	Protein

▪ ▪ ▪

B. K-Nearest Neighbors (K-NN) Classification – Illustrations



Training Set ($n = 15$ entries)



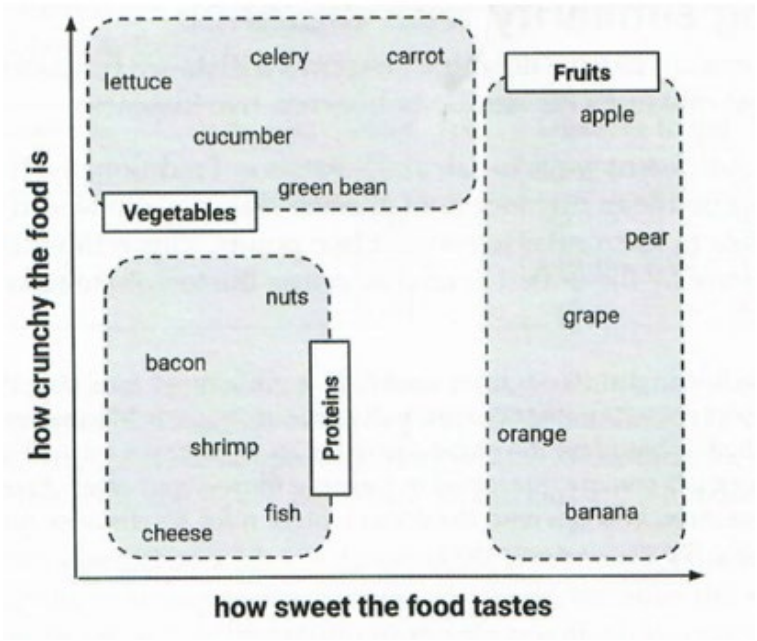
Test Set (in this case, 1 entry)

a. The object "tomato" will be assigned to one of the 3 classes, based on the class it is closest to, using some distance measure.

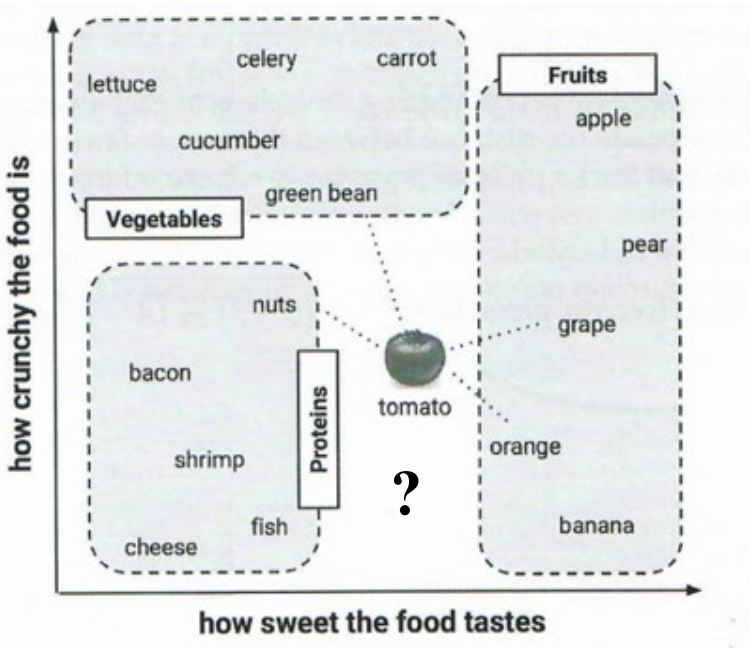
b. For K-NN, K is the number of nearest neighbors to use in the comparison. K is selected *a priori* (note: K has a different meaning than the k in k-Means clustering).

c. The distance from tomato to each labeled/known entry is calculated. Tomato is assigned to the class representing the plurality/most common of the K nearest neighbors (the plurality/most common of the K smallest distances).

B. K-Nearest Neighbors (K-NN) Classification – Illustrations



Training Set ($n = 15$ entries)



Test Set (in this case, 1 entry)

Classify tomato with K-NN algorithm for $K = 3$

For tomato: sweetness $= 6$, crunchiness $= 4$

Euclidean distance to 4 entries in training set

Ingredient	Sweetness	Crunchiness	Food type	Distance to the tomato
Grape	8	5	Fruit	sqrt((6 - 8)^2 + (4 - 5)^2) = 2.2
Green bean	3	7	Vegetable	sqrt((6 - 3)^2 + (4 - 7)^2) = 4.2
Nuts	3	6	Protein	sqrt((6 - 3)^2 + (4 - 6)^2) = 3.6
Orange	7	3	Fruit	sqrt((6 - 7)^2 + (4 - 3)^2) = 1.4

3 nearest neighbors:
2 are fruits, 1 is protein

Prob(T=fruit) $= 2/3$

Prob(T=protein) $= 1/3$

C. Selecting the number K – some comments

- As we saw, the value of K (the number of nearest neighbors) used can influence the classification results.

- Suppose $K = n$, the total number of entries in the training set. Then a test value will always be assigned to the class with the most entries in the training set, regardless of its nearest neighbors.

- Suppose $K = 1$. Then a test value will always be assigned to the same class as the entry in the training set it is closest to. This allows noisy data and outliers to greatly influence the classification.

- In practice, there are guidelines for selecting K (e.g., $K = \sqrt{n}$, use several different K values, etc.). With large and balanced data sets, selection of K is less critical.

D. Data scaling

Features are generally scaled/transformed/normalized to a standard range before applying the K-NN algorithm. This allows for features with different scale ranges to be used simultaneously.

In the food example, if we added a 3rd attribute of spiciness, which is measured using the Scoville scale ranging from 0 (not spicy) to over 10^6, the spiciness attribute would dominate the distance calculation.

Min-Max scaling

$$w_{scaled} = \frac{w - \min(w)}{\max(w) - \min(w)}$$

Standard Normal (z-score) scaling

$$w_{scaled} = \frac{w - \overline{w}}{s_w}$$

w – vector of values of a feature / attribute in X (dim n)
$\min(w)$ – smallest value in the vector (dim 1)
$\max(w)$ – largest value in the vector (dim 1)
$\overline{w}, s_w$ – mean and std. dev. of elements of vector w (each: dim 1)
w_{scaled} – scaled vector of values of a feature / attribute in X (dim n)

Note: categorical attributes (both nominal and ordinal) can also be used in K-NN

E. Example: Quality Control of Surgical Mesh

Physical testing of surgical mesh includes two attributes:

- Strength of material (measured in kg/m^2)
- Flexibility (measured on an integer scale from 1 to 10; minimum to maximum amount of flexibility)

Assessment by 100 surgeons

- Products are evaluated by surgeons who give a Pass/Fail grade

Surgical mesh sourced from 4 different vendors was tested and assessed by surgeons with the following results (training set):

Vendor	x1: strength (kg/m^2)	x2: flexibility (1-min to 10-max)	y: Classification by surgeons
A	7	7	Pass
B	4	7	Pass
C	4	3	Fail
D	4	1	Fail

Mesh from a new vendor was tested, resulting in $x1 = 7$ and $x2 = 3$. Without doing an expensive evaluation by surgeons, classify the new vendor mesh as Pass or Fail.

Use a K-NN classification with $K = 3$

Calculate the distances between the attributes of the new vendor's mesh to each of the meshes in the training set

Vendor	x1	x2	Distance to (7,3)
A	7	7	$\sqrt{(7-7)^2+(7-3)^2}=4$
B	4	7	$\sqrt{(4-7)^2+(7-3)^2}=5$
C	4	3	$\sqrt{(4-7)^2+(3-3)^2}=3$
D	4	1	$\sqrt{(4-7)^2+(1-3)^2}=3.61$

Rank order the distances of the new mesh
Identify the $K = 3$ nearest neighbors

Vendor	x1	x2	Distance to (7,3)	Rank Order	$K = 3$ nearest neighbor?
A	7	7	4	3	Yes
B	4	7	5	4	No
C	4	3	3	1	Yes
D	4	1	3.61	2	Yes

Indicate category (Pass/Fail) of $K = 3$ nearest neighbors

Vendor	x1	x2	Distance to (7,3)	Rank Order	$K = 3$ nearest neighbor?	Category of nearest neighbor
A	7	7	4	3	Yes	Pass
B	4	7	5	4	No	--
C	4	3	3	1	Yes	Fail
D	4	1	3.61	2	Yes	Fail

Classify new mesh as Pass or Fail:

Probability of 2/3 that new mesh will Fail the assessment by surgeons